Symbiosis Institute of Technology

Faculty of Engineering

CSE- Academic Year 2025-26

Internet of Things Lab Batch 2023-27

Lab Assignment No: - 9	
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Batch	C1
Class	CSE
Academic Year & Semester	6TH Semester
Title of Assignment:	Design and implementation of an IoT-based system for collecting sensor data and transmitting it to a cloud platform (Firebase Realtime Database) at regular intervals of 15 minutes using Wi-Fi

THEORY:

Aim: To design and implement an IoT based system for collecting sensor data & transmitting it to a cloud platform (Firebase) at regular intervals of 15 secs

Theory: ~~The IoT~~ The ESP8266 functions as the primary processing and communication unit. It is an economical microcontroller equipped with built-in Wi-Fi capabilities, which makes it highly suitable for IoT applications. The device collects data from connected sensors and transmits it to cloud-based platforms via the internet.

The DHT11 sensor is used to measure the environmental parameters such as temperature & humidity. It provides digital output signals, enabling easy interfacing with the microcontroller. The sensor periodically captures and sends the measured data to the ESP8266 for further processing. The processed data is then uploaded to the Firebase Realtime Database, a cloud-based NoSQL database service. This database stores information in JSON format & supports real-time data synchronization, allowing users to access and monitor the data instantly from any location.

To ensure efficient operation, the system uses time-based ~~monitoring~~ scheduling.

Connections:

DHT11	ESP8266
VCC	V3V
GND	GND
Data	D4

ALGORITHM:

Algorithm:

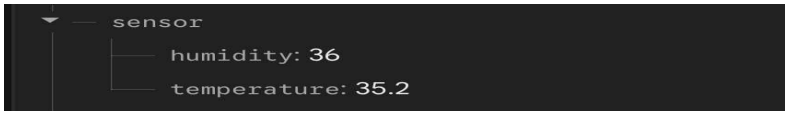
1. Start
2. Initialize DHT11 sensor
3. Initialize wifi module
4. Connect to wifi network
5. Configure Firebase
6. Read temperature & humidity data
7. Upload data to Firebase
8. Wait for 15 sec

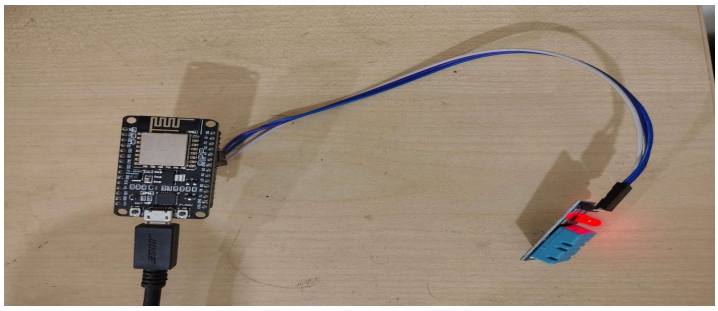
CODE:

```
#include <WiFi.h>
#include <Firebase_ESP_Client.h>
#include  <DHT.h>

// WiFi credentials
#define WIFI_SSID "Swarali's A35"
#define WIFI_PASSWORD "12345678"

// Firebase credentials
#define API_KEY "AIzaSyAlaT6UNIlMaSXRXTfpesIevkr8MEQ_RQ8"
#define DATABASE_URL "iotlab1-b71e1-default-rtdb.firebaseio.com"

#define DHTPIN 4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

FirebaseData fbdo;
FirebaseAuth auth;
FirebaseConfig config;

void setup() {
  Serial.begin(115200);

  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  Serial.print("Connecting");

  while (WiFi.status() != WL_CONNECTED) {
    Serial.print(".");
    delay(500);
  }
  Serial.println("\nConnected!");

  // Firebase config
  config.api_key = API_KEY;
  config.database_url = DATABASE_URL;

  config.signer.test_mode = true;

```

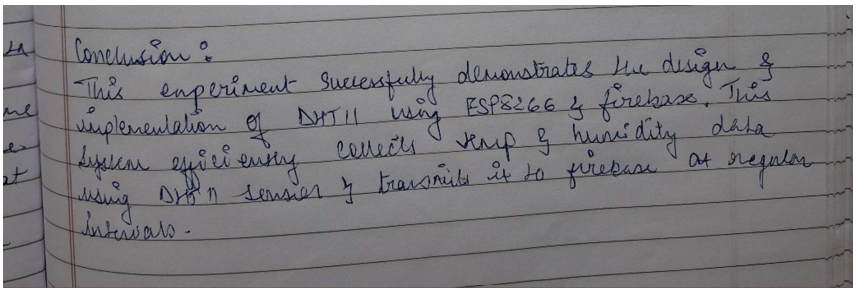
```
  Firebase.begin(&config, &auth);
  Firebase.reconnectWiFi(true);

  delay(2000); // ⭐ IMPORTANT (stability)

}

void loop() {
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();

  // Check sensor
  if (isnan(temp) || isnan(hum)) {
    Serial.println("❌ Failed to read from DHT");
    delay(2000);
    return;
  }

  Serial.print("Temp: ");
  Serial.println(temp);
  Serial.print("Humidity: ");
  Serial.println(hum);

  // Send to Firebase
  if (Firebase.RTDB.setFloat(&fbdo, "/sensor/temperature", temp)) {
    Serial.println("Temp sent");
  } else {
    Serial.println("Error: " + fbdo.errorReason());
  }

  if (Firebase.RTDB.setFloat(&fbdo, "/sensor/humidity", hum)) {
    Serial.println("Humidity sent");
  } else {
    Serial.println("Error: " + fbdo.errorReason());
  }

  delay(5000); // every 5 sec
}
```

OUTPUT:





CONCLUSION:



Conclusion:
This experiment successfully demonstrates the design & implementation of DHT11 using ESP8266 & firebase. This system efficiently collects temp & humidity data using DHT11 sensor & transmits it to firebase at regular intervals.